Serial No: 101964

Registration No: CIIT/SP19-BSM-041/ISB

COMSATS University Islamabad

SAAD-UR-REHMAN s/o AZIZ-UR-REHMAN

Of

Islamabad Campus

has been conferred upon the degree of

Bachelor of Science in Mathematics

Given on this Fifteenth day of March two thousand and Twenty Three at Islamabad

Date of Issuance: 15th March 2023

Controller of Examinations

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04 AUG 2023
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Islamabad

Prepared By: .....

Checked By: .....

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Admission Letter – Academic cycle 2024/2026
Joint MSc programme "MathMods"

To: Mr Saad Ur REHMAN
saadurrehman1100@gmail.com
Dhoke Chaudhrian Choke Cob Lane Rawalpindi Cantt 949/14-A-1
46000 Rawalpindi (Pakistan)

L'Aquila, Saturday, 18 May 2024

Dear **Saad Ur REHMAN**,

Born in Pakistan on 05/03/2001, Passport no. CS5428142,

This is to inform you that your application for the two-year Joint Master's Programme "Mathematical Modelling in Engineering" (*MathMods*) has been positively evaluated and your name has been included in the list of eligible students who will be awarded a scholarship worth 7,000 EUR, as detailed below.

We have received around 1340 applications, many of which from extremely talented students, but only those evaluated with a score equal to or greater than 66/100 (as is your case), have been considered eligible for the Programme. Therefore, we are pleased to confirm that you have been admitted to the 2024 edition of the *MathMods* Joint MSc starting next September.

PARTIAL FEE WAIVER AND PARTICIPATION COSTS

Based on your position in our final absolute ranking list, the *MathMods Consortium* has established to grant you a partial waiver of the participation costs. In line with that, you will be required to pay a **total of 2,000 EUR** only (500/semester) for participation costs, instead of the standard EUR 9,000.

Note that participation costs include the tuition fees as well as any other mandatory costs related to the student's participation in the whole two-year programme. However, they do not include insurance, accommodation costs, living costs, travel costs, books or other related expenses.

Participation costs are due in five instalments, as follows:

- **EUR 344** as the initial non-refundable fee, which you **have already paid**.
- EUR 156 is to be paid before the beginning of the first semester.
- EUR 500 is to be paid at the beginning of each subsequent semester, for a total of EUR 1,500 in three instalments.

DATES AND MOBILITY PATH

You must begin the MathMods Programme on 23 September 2024 and are expected to complete it by 22 September 2026.

As you know, our Consortium is coordinated by the **University of L'Aquila** in Italy, and involves two leading institutions in Europe: the **University of Côte d'Azur** (France), and the **University of Hamburg** (Germany).

The *MathMods* Master Course will be structured in four semesters. Each semester will award 30 ECTS credits. The mobility scheme will involve two different locations, at least. More specifically,

1) All students will spend their **first semester** (23 September 2024 – 31 March 2025) in **Italy** (University of L'Aquila).

2) All students will spend their **second semester** (1 April 2025 – 31 August 2025) in **Germany** (University of Hamburg). Students will attend some introductory activities related to the second-semester subjects during March at the University of L'Aquila (Italy).

3) All students will spend their second year in one of the three universities within the Consortium. The *MathMods* Consortium will inform each student about the details of their mobility path (point 3) before the beginning of the second semester.

Further details on the Master Course can be found on the *MathMods* website: www.mathmods.eu

PRE-MASTER FOUNDATION PROGRAMME

Although Mathematics has a universal language, it is taught in many ways, depending both on the geographical area and the main undergraduate subject (pure maths, applied maths, engineering, physics, computer science). It is important that you start your MathMods career with all the needed background to tackle the Semester 1 subjects, which are strongly focused on theoretical mathematics. To help you in this task, the University of L'Aquila offers you free of charge a "Pre-Master's Foundation Programme in Applied Mathematics" **entirely taught remotely** as scheduled below. Please note the Pre-Master's Programme is only reserved for students who did not graduate from a university based in Italy.

- The first part starts in the second half of July and will be over before the beginning of the Master's activities (15 July - 15 September).
- The second one takes place at the beginning of Semester 2 but is not meant for MathMods students.

The Pre-Master modules addressed to MathMods students are the following (all being delivered in the first part of the programme):

A. Real Analysis: Foundations (set theory, real numbers, complex numbers) - 6 ECTS.

B. Differential Equations: Foundations (linear and non-linear first-order differential equations, linear second-order differential equations) - 3 ECTS.

C. Introduction to Linear Algebra (linear spaces, matrices, etc) - 3 ECTS.

D. Italian for Absolute Beginners – 3 ECTS.

There is a dedicated page, which we invite you to check to obtain further information and updates: *https://www.intermaths.eu/pre-master-foundation*

However, unlike other international students, the Pre-Master Foundation Programme follows a slightly different scheme for MathMods students, as explained below:

1. After enrolling in the Pre-Master, a student may decide to attend one or more of the above modules. One may even decide to attend none of them, either.

2. At the end of each Pre-Master module, students sit an online exam, which can be taken on two dates: one at the end of the module, and another one before the beginning of the Master's activities.
3. Although attendance is not compulsory for MathMods students, you **must sit the Pre-Master's exams**:
    a. If the exam is passed, you will have the corresponding ECTS credits awarded on your Pre-Master's transcript, which is completely detached from your MathMods MSc transcript.
    b. If the exam is not passed, that will have no repercussions whatsoever on your MathMods career.

   In either case, the MathMods consortium will keep track of their performance in their internal records.
4. Lastly, note that by enrolling in the Pre-Master Programme, you immediately obtain a University of L'Aquila email account (@student.univaq.it), access to the Office365 suite (which is used throughout the first semester to complement the teaching activities and to interact with your lecturers and colleagues), not to mention immediate access to our university wi-fi networks. Having such an account will greatly help you also during the forthcoming summer period, for example, to attend induction activities. Our staff will assist you with that.

COMMUNICATION, ATTENDANCE, EXAM, ETC.

The information below mostly refers to your first semester at the University of L'Aquila. Similar rules are applied in the following semesters, as well. However, they will be communicated in detail by the related coordinator of the partner institution.

As students at the University of L'Aquila, all students are provided with an institutional email address name.surname@student.univaq.it. All communications from/to Lecturers and Staff must be carried out through said email address. The MathMods Staff uses the email address *mathmods@univaq.it* to communicate with the students, individually and jointly depending on the subject. Emails from the MathMods Staff range from general communications (lectures schedule, exams, events, etc) to specific communication to a student (for example a required document about previous qualifications), from instructions on specific tasks (such as attendance sheets) to official notifications, from instructions on how to apply for additional funding to reminders on specific tasks. **All students are responsible for their frequent and careful check of the emails coming from the MathMods Staff.**

Attendance of all courses and mid-term tests is compulsory. Attendance should start at the scheduled start time and end at the scheduled end time. Attendances will be verified via online forms and/or attendance sheets and/or via direct checks by the lecturers. All **absences** should be motivated by a valid reason and communicated **in advance, if possible,** to the email address mathmods@univaq.it. In case of repeated unmotivated or unsuitably motivated absences, or in case of other disciplinary issues that may arise, the MathMods Coordinator reserves the right to formally put the student on probation and, in case

of failure to return to good standing after the probation period, to exclude the student from the programme.

Students are required to pass all exams of a given semester within the examination session scheduled at the end of the semester and before the beginning of the subsequent semester. Students failing to accomplish this task will lose the status of MathMods students. If a student fails to pass one or more exam(s) at the end of a given semester, the MathMods Consortium reserves the right, in case of evidence of external factors/impediments such as serious health conditions or force majeure, to allow the student to remain in the programme and to set up a re-sit plan. If the student loses the status of MathMods student, the University of L'Aquila may propose the student switch to other study programmes in applied mathematics.

INSURANCE

All *MathMods* students must hold valid insurance covering risks related to health, accidents, death, permanent invalidity, and civil responsibility for the whole duration of the Programme. An insurance card will also be required by local authorities to apply for a residence permit (if applicable).

VISA AND RESIDENCE PERMIT

If you need a **visa** to stay in Europe, you will have to submit a pre-enrolment application through a dedicated website: *www.universitaly.it* (our staff will contact you shortly to provide step-by-step directions on this matter). Once your application to Universitaly has been approved by the University of L'Aquila, you will have to get in contact with the Italian Diplomatic-consular mission to start the visa request process.
Please note that, before arriving in Europe next September, you will only need to apply for a visa at the local Italian Embassy/Consulate, as Italy is the location for your Semester 1. You will normally receive an Italian visa valid for one year (notice that it is necessary to have a passport valid until 3 months after the expiry of your visa). Afterwards, during your Semester 1 you will have to apply for German visa (as Germany will be the location for your Semester 2) at the Germany Consulate located in Rome, and so on.

Regarding the Italian visa, we remark that the Italian Embassy/Consulate in your country may ask you to provide a certificate giving evidence that you are competent in Italian (corresponding at least to the B1 level of the Council of Europe). However, be informed that you will receive an Italian visa even without such a certificate, as the *MathMods* MSc is entirely taught in English. It will, nevertheless, be compulsory for you to acquire a basic level of proficiency in the Italian language (A1, according to C.E.F.R.) before the end of your Semester 1. Moreover, if you spend your third and/or fourth semester in L'Aquila, you will also have to pass a secondary test of competence in Italian (A2 level, according to C.E.F.R.) before the end of your third/fourth semester there. Similar courses will also be offered in the other partner institutions.
If you need a visa to stay in Italy, remember that you will also have to **apply for an Italian residence permit within 8 days of arriving in our country.** A residence permit is indeed required when a foreigner stays in

Italy for longer than three months. The *MathMods* team will assist you in preparing all the required documentation.

ENROLMENT

Be aware that, no matter your citizenship, if you gained your **previous qualification** (e.g., bachelor's degree) outside Italy, you will have to provide **one** of the following certificates to finalise your enrolment at the L'Aquila Registrar's office:

- **CIMEA** (www.cimea.it) Statement of comparability – follow the dedicated link https://cimea.diplome.eu/univaq/#/auth/login (after registration) to take advantage of a discounted fare available to prospective students of the University of L'Aquila.
- **Declaration of Value (DoV)** – to be requested from the Italian diplomatic authorities operating in the country where you gained your previous qualification.
- **Diploma Supplement (DS)** – note that you may submit a Diploma Supplement to certify your previous qualification only if you gained your undergraduate degree from an institution issuing Diploma Supplements.

In place of the previous certificates, you may entitle the University of L'Aquila to contact the institution where you obtained your bachelor's degree and verify the genuine nature of your qualification. For this purpose, you must provide details of an administrative responsible person, typically working in the registrar's office, who must send a confirmation letter upon request from the University of L'Aquila to verify that you have obtained your bachelor's degree from that institution and that your final transcript and degree certificate are authentic. Further information about this procedure will be communicated via email from our staff.

Lastly, be informed that if you are applying for a visa, the local Diplomatic-consular mission might still require that you obtain a Declaration of Value (DoV) before they issue a study visa for you even though our university accepts other types of verifications, as explained above. If this is the case, you will have to go through the DoV request process, which can be quite lengthy, instead of the other options available at the University of L'Aquila. Therefore, make sure you start the whole process in good time in advance.

HOUSING FACILITIES

Please note that currently, the University of L'Aquila has not got its own halls of residence, though there is a private one recently opened in the city centre. We thus suggest you opt for renting a flat from private owners, the cost per person ranging from 200 to 350 euros per month depending on several factors (e.g., shared or private rooms, utilities included or not, area). You will be assisted in this regard by the "Aquilasmus - ESN L'Aquila" Association, an official member of the Erasmus Student Network (URL: www.esnlaquila.it, address: c/o University of L'Aquila, Palazzo Camponeschi - Piazza Santa Margherita, 2 - 67100 L'Aquila). The Aquilasmus staff will meet you within a few days after your arrival to help you find a place to stay for the whole term. For further information, including possible solutions for temporary

accommodation on your arrival, we invite you to visit this web page: https://www.mathmods.eu/apply/admitted-students/accommodation

DISCLAIMER

All MathMods students must refrain from discussing any political, geopolitical, or religious matter during the MathMods joint activities.

Finally, we suggest you join the MathMods & InterMaths Alumni Association (URL: www.mathmods.eu/alumni, address: c/o DISIM - via Vetoio snc - 67100 L'Aquila). The mission of the Association is to serve the interests of our Students and Alumni providing support with applying for regional grants, residence permits, travel/health insurance, etc.). The Association will contact you in August to provide information about their services and how to become a member.

For any enquiries, please do not hesitate to contact our team at *mathmods@univaq.it*
We are looking forward to meeting you in Italy on September 23!

Best Regards

Prof. Bruno Rubino
Vice-Rector for International Affairs

DISIM - Department of Information Engineering,
Computer Science and Mathematics
University of L'Aquila, Via Vetoio (Coppito)
67100 L'Aquila (AQ) – Italy
Email: mathmods@univaq.it, URL: www.mathmods.eu

---------- STUDENT'S SECTION ----------

I acknowledge that I have read and fully understand the above information and that I will comply with the same.

Date: Saturday, 18 May 2024 **Signature:** _____

Saad Ur REHMAN

Dipartimento di Ingegneria e Scienze dell'Informazione e Matematica
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

CERTIFICATE OF ATTENDANCE

We certify that ***Saad Ur Rehman*** *a*ttended the Summer School and Conference "***Mathematics for Signal Processing and Applications in Geophysics and other fields (MaSAG)"*** *L'Aquila, 9-14 June, 2025. The event lasted 8 hours each of the three days of the School and 6 hours each of the three days of the Conference, for a total of 42 hours.*

L'Aquila, June 17, 2025

Prof. Antonio Cicone
SCHOOL DIRECTOR

Prof. Antonio Cicone
Associate Professor of Numerical Analysis
email: antonio.cicone@univaq.it
web: http://people.disim.univaq.it/~antonio.cicone/

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

DISIM
Dipartimento di Ingegneria e Scienze dell'Informazione e Matematica

POLITECHNIKA GDAŃSKA

Gdańsk, 08.09.2025 r.

ZAŚWIADCZENIE

o przyjęciu cudzoziemca na studia stacjonarne

Zaświadcza się, że Pan **Saad Ur Rehman**
urodzony dnia 05-03-2001 w Rawalpindi
posiadający obywatelstwo Pakistan
legitymujący się dokumentem tożsamości paszport CS5428142
wydanym w Pakistan
został przyjęty* na okres od 01.10.2025 r. do 30.09.2025r.
na studia: Kierunek: Nanotechnology (joint Master's double-degree program) (WFTiMS), II stopnia, stacjonarne, 2025/2026 - zimowy
na zasadach **bez odpłatności** na pierwszy semestr.

Kształcenie na studiach stacjonarnych rozpoczyna się w dniu 01.10.2025 r.
Studia stacjonarne odbywają się w języku **angielskim**,
na poziomie biegłości językowej **B2**.

Nazwa programu finansowanego przez Unię Europejską lub przez państwa członkowskie Unii Europejskiej, który promuje przemieszczanie się obywateli państw trzecich w Unii Europejskiej lub w państwach członkowskich Unii Europejskiej uczestniczących w tym programie lub porozumienia między przynajmniej dwiema instytucjami szkolnictwa wyższego przewidującego mobilność wewnątrzunijną: ***RealMaths Implementation Agreement***
oraz nazwy i adresy jednostek prowadzących studia stacjonarne albo kształcenie w szkole doktorskiej, objętych tym porozumieniem: ***Politechnika Gdańska, ul. Narutowicza 11/12, 80-233 Gdańsk oraz University of L'Aquila, Piazza Santa Margherita 2, 67100 L'Aquila, Italy.***
Informacja o dotychczasowej realizacji toku studiów stacjonarnych i zaliczeniu wymaganych programem studiów przedmiotów/ kształcenia w szkole doktorskiej i zaliczeniu wymaganych programem kształcenia przedmiotów: ***nie dotyczy***

POLITECHNIKA GDAŃSKA
CENTRUM WSPOŁPRACY MIĘDZYNARODOWEJ
ul. G. Narutowicza 11/12, 80-233 Gdańsk
NIP 584 020 35 93 REGON 000001625
[illegible]

Prorektor ds. studenckich
dr Barbara Wikieł, prof. uczelni

.....
Prorektorka ds. studenckich
dr Barbara Wikieł, prof. uczelni
(podpis kierownika jednostki prowadzącej studia stacjonarne)

* Osoba przyjęta na studia pełnię praw studenta otrzymuje po spełnieniu wymogów art. 83. ustawy z dnia 20 lipca 2018 roku - Prawo o szkolnictwie wyższym i nauce (Dz. U. z 2022. poz. 574 z późn. zm.).

POLITECHNIKA GDAŃSKA

Gdańsk, 08.09.2025

CERTIFICATE

of admission of a foreigner to full-time studies

It is hereby certified that Mr **Saad Ur Rehman**
born on 05-03-2001, in Rawalpindi
Citizenship Pakistan
ID card number passport CS5428142
issued in Pakistan
was accepted* for the period from 01.10.2025 till 30.09.2025
for studies: Field of study: Nanotechnology (joint Master's double-degree program) (WFTiMS), second-cycle studies, Full-time studies, 2025/2026 - winter
on a non-paying basis for the first semester.

Full-time education begins on 01.10.2025
Full-time studies are held in **English**,
at the language level **B2**.

Name of the program financed by the European Union or by Member States of the Union which promotes the mobility of third-country nationals within the European Union or in the Member States of the European Union participating in this program or the agreement between at least two higher education institutions providing for intra-EU mobility: ***RealMaths Implementation Agreement***
and the names and addresses of units providing full-time studies or doctoral school education, covered by this agreement: ***Politechnika Gdańska, ul. Narutowicza 11/12, 80-233 Gdańsk, Poland and University of L'Aquila, Piazza Santa Margherita 2, 67100 L'Aquila, Italy***.
Information on the current implementation of full-time studies and the required credits for subjects in the curriculum / education at the doctoral school and the required credits for subjects in the curriculum: ***not applicable***

Vice-Rector for Student Experience
Professor Barbara Wikieł

POLITECHNIKA GDAŃSKA
CENTRUM WSPOLPRACY MIĘDZYNARODOWEJ
ul. G. Narutowicza 11/12, 80-233 Gdańsk
NIP 584-020-35-93 REGON 000001627
(2)

.....
Vice-Rector for Student Experience
Ph.D. Barbara Wikieł, Gdańsk Tech professor
(signature of the head of the unit running full-time studies)

* A person admitted to studies receives full student rights after meeting the requirements of Art. 83 of the Act of 20 July 2018 - Law on Higher Education and Science (Journal of Laws of 2022, item 574, as amended).

Oscillation of Ar_2Molecule

AKINWUMI, Akinsola Joel[1]

[1] Gdansk University of Technology, Poland

Abstract

This study investigates the interatomic interactions of two argon atoms separated by a distance d using the Lennard-Jones potential. Using LAMMPS Using the LAMMPS Molecular Dynamics Simulator and analytical methods, we explored energy interactions, equilibrium bond lengths, and frequency dispersion curves. By computing the oscillation of frequency f in different energy states $(-\epsilon, 0)$. The results of the study showed that as the energy increases, the frequency decreases, demonstrating harmonic behavior.

Keywords: Lennard-Jones potential, Argon gas, Molecular Dynamics, LAMMPS, equilibrium bond lengths, frequency dispersion curves

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1 Introduction

The interaction between atoms and molecules is a cornerstone of understanding physical and chemical processes at both macroscopic and microscopic scales. Among the models developed to describe such interactions, the Lennard-Jones potential, introduced by Sir John Lennard-Jones in 1924, is one of the most influential [1]. This potential provides a mathematical framework for quantifying the forces between electrically neutral atoms or molecules, particularly noble gases, which exhibit weak intermolecular forces. It combines attractive and repulsive components to describe the energy profile as a function of atomic separation.

The Lennard-Jones potential is widely used because of its simplicity and effectiveness in modeling van der Waals interactions. The repulsive term is attributed to the Pauli exclusion principle, which prevents particles from overlapping by prohibiting identical quantum states. Conversely, attractive terms model interactions such as London dispersion forces arising from instantaneous dipole-induced dipole interactions [2].

This study focuses on applying the Lennard-Jones potential to describe the interactions of argon gas atoms. Using both analytical methods and molecular dynamics simulations with LAMMPS [3], we investigate key properties such as equilibrium bond length, energy of dissociation, and force interactions. Furthermore, we validate the numerical results with theoretical predictions and explore the implications for atomic-scale behavior under varying conditions.

2 Related Work

The study of interatomic interactions using the Lennard-Jones potential has been extensively explored. Lennard-Jones' seminal work [1] introduced the potential, laying the foundation for modeling van der Waals interactions. Molecular dynamics simulations, such as those facilitated by the LAMMPS simulator [3], have provided deeper insights into atomic behavior. Studies like [4] focus on noble gases, including argon, showcasing the potential's utility in predicting equilibrium bond lengths and vibrational dynamics.

These works form the basis for our study, which extends this approach to analyze energy-frequency relationships under varying conditions.

3 Methodology

This study investigates the interaction between Argon atoms using the Lennard-Jones potential through analytical derivations and numerical simulations. The equilibrium bond length, dissociation energy, and

force of interaction are calculated, and results are validated with molecular dynamics simulations using LAMMPS. The methodology is divided into analytical computations and the use of simulation tools, with emphasis on deriving critical parameters and comparing their trends [4].

3.1 Equilibrium Bond Length

The equilibrium bond length is the separation distance at which the potential energy between two interacting atoms is minimized. Starting from the Lennard-Jones potential:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right], \tag{1}$$

we calculate the equilibrium bond length by finding the value of r where the derivative of $V(r)$ equals zero:

$$V'(r) = 4\epsilon \left[-12 \left(\frac{\sigma^{12}}{r^{13}}\right) + 6 \left(\frac{\sigma^{6}}{r^{7}}\right) \right] = 0. \tag{2}$$

Simplifying (2) yields:

$$-12 \left(\frac{\sigma^{12}}{r^{13}}\right) + 6 \left(\frac{\sigma^{6}}{r^{7}}\right) = 0 \quad \Rightarrow \quad r^6 = 2\sigma^6. \tag{3}$$

From (3), the equilibrium bond length is:

$$r_{\text{eq}} = 2^{\frac{1}{6}}\sigma. \tag{4}$$

At $r = r_{\text{eq}}$, the potential energy reaches its minimum value, V_{min}. Substituting r_{eq} back into (1), we find:

$$\begin{aligned} V_{\text{min}} &= 4\epsilon \left[\left(\frac{\sigma}{2^{\frac{1}{6}}\sigma}\right)^{12} - \left(\frac{\sigma}{2^{\frac{1}{6}}\sigma}\right)^{6} \right] \\ &= 4\epsilon \left[\frac{1}{4} - \frac{1}{2} \right] = -\epsilon. \end{aligned} \tag{5}$$

Thus, the equilibrium bond length r_{eq} and the corresponding minimum potential energy V_{min} provide foundational parameters for further computations and simulations.

3.2 Energy of Dissociation

The energy of dissociation refers to the amount of energy required to separate two interacting atoms completely, moving them to an infinite distance where the potential energy becomes zero. Starting from the Lennard-Jones potential [3], we compute the dissociation energy as the difference between the potential energy at equilibrium, V_{min}, and the potential energy as $r \to \infty$.

At infinite separation, the Lennard-Jones potential approaches zero:

$$\lim_{r\to\infty} V(r) = \lim_{r\to\infty} 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right] = 0. \tag{6}$$

Thus, the energy of dissociation is given by:

$$\begin{aligned} E_{\text{diss}} &= V_{\text{min}} - \lim_{r\to\infty} V(r) \\ &= -\epsilon - 0 = -\epsilon. \end{aligned} \tag{7}$$

This result shows that the dissociation energy is equal in magnitude to the depth of the potential well, ϵ, but with a negative sign. It represents the energy needed to overcome the attractive forces and completely separate the atoms.

Physically, the dissociation energy highlights the stability of the bond between two atoms. For Argon, this value is directly proportional to ϵ, which depends on the material properties and intermolecular forces.

3.3 Description of Simulation

The Lennard-Jones potential was simulated using the LAMMPS Molecular Dynamics Simulator to investigate the interaction between Argon atoms. The simulation involves varying the energy levels and calculating the corresponding atomic separations.

The energy levels, denoted as E_i, are defined incrementally based on the dissociation energy ϵ:

$$E_i = -\epsilon + i\Delta E, \quad \Delta E = \frac{\epsilon}{11}, \quad i = 0, 1, 2, \ldots, 10. \tag{8}$$

Here, $E_0 = -\epsilon$ and $E_{11} = 0$, representing the lowest and highest energy states, respectively. For each energy level E_i, we calculate the corresponding values of r by solving the Lennard-Jones potential equation (1).

To simplify the computation, we rewrite the Lennard-Jones potential as a quadratic equation in terms of $x = \left(\frac{\sigma}{r}\right)^6$:

$$E_i = 4\epsilon \left[x^2 - x \right]. \tag{9}$$

This corresponds to the general quadratic equation:

$$ax^2 + bx + c = 0, \tag{10}$$

where $a = 4\epsilon$, $b = -4\epsilon$, and $c = E_i$. The solutions for x are obtained using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{11}$$

Substituting a, b, and c into (11), we find:

$$x = \frac{4\epsilon \pm \sqrt{16\epsilon^2 + 16\epsilon E_i}}{8\epsilon}. \tag{12}$$

For each value of x, the corresponding r values are calculated as:

$$r = \frac{\sigma}{x^{1/6}}. \tag{13}$$

Two possible solutions for r are obtained from each energy level, corresponding to the two branches of the potential curve (one for attraction and one for repulsion). Among these, the larger value represents the stable configuration of the Argon atoms.

The simulation begins with these initial separations r as the bond lengths and evolves over time using a time step of $h = 0.5\,\text{fs} = 0.5 \times 10^{-15}\,\text{s}$. The output data include the potential energy, bond length, and other relevant properties, which are later analyzed and compared with theoretical predictions.

3.4 Force Calculation

The Lennard-Jones potential is a classical model describing the interaction between neutral atoms or molecules [1]. The force between two interacting atoms can be derived by taking the negative gradient of the potential, which gives the interaction force as a function of the separation distance r.

Starting from the Lennard-Jones potential $V(r)$ given in (1), the force is expressed as the negative derivative of the potential:

$$F(r) = -\frac{dV(r)}{dr}. \tag{14}$$

Taking the derivative of the potential $V(r)$ with respect to r, we obtain:

$$F(r) = -4\epsilon \left[-12 \left(\frac{\sigma^{12}}{r^{13}} \right) + 6 \left(\frac{\sigma^6}{r^7} \right) \right]. \tag{15}$$

Simplifying (15), the force becomes:

$$F(r) = 24\epsilon \left[\frac{2\sigma^{12}}{r^{13}} - \frac{\sigma^6}{r^7} \right]. \tag{16}$$

This expression describes the force between two Argon atoms at a given separation distance r.

At the equilibrium bond length $r = r_{\text{eq}}$, we can substitute the equilibrium separation into (16). Since the force at equilibrium is expected to be zero (as the atoms are at their most stable configuration), we find:

$$F(r_{\text{eq}}) = 0. \tag{17}$$

Thus, at the equilibrium separation r_{eq}, the attractive and repulsive forces exactly balance each other, resulting in no net force between the atoms. The behavior of the force around equilibrium can be studied to understand how the force changes as the atomic separation increases or decreases, which is essential for understanding atomic interactions in real systems [3].

In the simulation, we use the calculated force values to predict how the atoms behave as they interact, plotting force as a function of separation distance to visualize these interactions.

3.5 Classical computation of the force

In this section, we analyze the force between two Argon atoms using a classical approach, assuming small oscillation or small deviation from the equilibrium. We assume that the two atoms are connected by a spring with spring constant K (N/m). The law is described by Hooke's law:

$$F = -Kx \tag{18}$$

Where F is the force of compression or extension in our study, compression and extension are analogous to attraction and repulsion, respectively, and x is the displacement from equilibrium.

Now, how does the spring constant relate to the potential of Lennard-Jones? We note that Lennard-Jones gave the approximate potential between interacting atoms, we can expand this potential about the equilibrium bond length r_{eq} with Taylor's series thus:

$$f(x) = \sum_{i=0}^{\infty} \frac{1}{i!} f^{(i)}(a)(x-a)^i \tag{19}$$

$$V(r) = \sum_{i=0}^{\infty} \frac{1}{i!} V^{(i)}(r_{\text{eq}})(r - r_{\text{eq}})^i \tag{20}$$

Approximating only to the second order, we have:

$$V(r) \approx \sum_{i=0}^{2} \frac{1}{i!} V^{(i)}(r_{\text{eq}})(r - r_{\text{eq}})^i \tag{21}$$

We will observe that at $i = 0$, $V(r) \approx V(r_{\text{eq}}) = V_{\text{min}} = -\epsilon$. Also, at $i = 1$, we have $V'(r_{\text{eq}})(r - r_{\text{eq}})$, recall from

equation (9), $V'(r) = -F(r)$, and from equation (10), we have that $F(r_{\text{eq}}) = 0$. Finally, at $i = 2$, we have:

$$\frac{1}{2}V^{(2)}(r_{\text{eq}})(r - r_{\text{eq}})^2$$

We note that the potential energy of a spring is given by:

$$V(x) = \frac{1}{2}Kx^2 \tag{22}$$

Where x is the displacement from the equilibrium, this is equivalent to $(r - r_{\text{eq}})$. Since $V(r_{\text{eq}})$ represents the potential at equilibrium and also since we are considering small oscillations, the value of $V(r_{\text{eq}})$ does not have any influence. Hence, on comparison of equation (13) with the results in equation (12), we obtain:

$$V = \frac{1}{2}Kx^2 = \frac{1}{2}V''(r_{\text{eq}})(r - r_{\text{eq}})^2$$

Thus:

$$K = V''(r_{\text{eq}}) \tag{23}$$

Now, we can compute $K = V''(r_{\text{eq}})$.

The second derivative of $V(r)$ is:

$$V'(r) = 4\epsilon\left[-12\left(\frac{\sigma^{12}}{r^{13}}\right) + 6\left(\frac{\sigma^6}{r^7}\right)\right] \tag{24}$$

$$V''(r) = 4\epsilon\left[12 \times 13\left(\frac{\sigma^{12}}{r^{14}}\right) - 6 \times 7\left(\frac{\sigma^6}{r^8}\right)\right]$$

Now, substituting $r = r_{\text{eq}}$, we have:

$$V''(r_{\text{eq}}) = \frac{4\epsilon}{\sigma^2}\left[12 \times 13\left(\frac{1}{r^{14}}\right)\sigma^{12} - 6 \times 7\left(\frac{1}{r^8}\right)\sigma^6\right]$$

Now, let's calculate K.

$$K = \frac{0.0416}{3.4^2}\left[\frac{156}{2^{\frac{7}{3}}} - \frac{42}{2^{\frac{4}{3}}}\right] \text{ (eV)}$$

$$= 0.0035986\,[30.954 - 16.6677]$$

$$= 0.0514\,\text{eV/Å}$$

The reduced mass M of two interacting particles (in this case, two Argon atoms) is given by the formula:

$$M = \frac{m_1 m_2}{m_1 + m_2}$$

Where:

m_1 and m_2 are the masses of the two particles (since both particles are Argon atoms, $m_1 = m_2 = m_{\text{Ar}}$).

For Argon:

The atomic mass of Argon $m_{\text{Ar}} = 39.948$, and $1\,eV/ = 1.660539 \times 10^{-27}$ kg.

So, the reduced mass M is:

$$M = \frac{m_{\text{Ar}}}{2} = \frac{39.948}{2} = 19.974\,\text{a.m.u}$$

Convert this into kilograms: The reduced mass M is given by:

$$M = 19.974 \times 1.660539 \times 10^{-27}$$

$$M \approx 3.315 \times 10^{-26}\,\text{kg}$$

The frequency f is related to the period of oscillation T by the formula:

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{M}}$$

Where: - $M = 3.315 \times 10^{-26}$ kg (reduced mass), - $K = 0.0514\,\text{eV/Å}^2$.

To convert the spring constant K to SI units, we use the following conversion factors:

$$1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}, \quad 1\,\text{Å} = 1 \times 10^{-10}\,\text{m},$$

So:

$$K_{\text{SI}} = 0.0514 \times \left(\frac{1.602 \times 10^{-19}}{(1 \times 10^{-10})^2}\right)\text{ N/m}^2.$$

Now, substitute the values into the formula for K_{SI}:

$$K_{\text{SI}} = 0.0514 \times \left(\frac{1.602 \times 10^{-19}}{1 \times 10^{-20}}\right)$$

$$K_{\text{SI}} = 0.0514 \times 1.602 \times 10^1\,\text{N/m}^2$$

$$K_{\text{SI}} = 0.0824\,\text{N/m}^2$$

Finally, we calculate the frequency f:

$$f = \frac{1}{2\pi}\sqrt{\frac{0.0824}{3.315 \times 10^{-26}}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{K_{\text{SI}}}{M}} = \frac{4.985 \times 10^{12}}{6.2832} \approx 7.93 \times 10^{11}\,\text{Hz}.$$

Thus, the frequency of oscillation is approximately:

$$f \approx 7.93 \times 10^{11}\,\text{Hz}.$$

This gives us the frequency value for the Argon atoms. Now to compute the wave number and period

$$\tilde{v} = \frac{f}{c} = \frac{7.93 \times 10^{11}}{3.0 \times 10^{10}}$$

$$\tilde{v} = 26.43\,\text{cm}^{-1}$$

$$T = \frac{1}{f} = \frac{1}{7.93 \times 10^{11}}$$

$$T = 1.26 \times 10^{-12}\,\text{s}$$

Thus: - The wave number is $\tilde{v} = 26.43\,\text{cm}^{-1}$, - The period is $T = 1.26 \times 10^{-12}$ s.

3.6 Calculation of the Dispersion Curve

The dispersion curve, which relates frequency (in cm^{-1}) to energy (in eV), is calculated using data generated from the LAMMPS simulation. The simulation provides a column of potential energy values in the output, corresponding to the initial bond length [3]. These potential energy values are plotted against time to identify the complete oscillation cycle, which returns to the initial potential energy value [2]. The period T of oscillation is recorded in femtoseconds.

To convert the recorded period T into seconds, it is multiplied by the time step of the simulation, 0.5×10^{-15} s. The corresponding frequency f is then calculated using the relation:

$$f(\text{cm}^{-1}) = \frac{1}{T(\text{s}) \times c},$$

where $c = 3.0 \times 10^{10}$ cm/s is the speed of light.

The calculated frequency values are plotted against energy to obtain the dispersion curve. This curve illustrates the inverse relationship between frequency and energy, reflecting the fact that as the bond length increases, the period T increases, resulting in a decrease in frequency f. This relationship is consistent with theoretical expectations, providing valuable insight into the vibrational behavior of the system.

The final dispersion curve clearly demonstrates how the frequency of vibration (in cm^{-1}) varies with potential energy (in eV), offering a visual representation of the system's dynamics.

4 Result

This section presents the numerical results obtained from the LAMMPS Molecular Dynamics Simulator. The results are visualized using GNUPLOT, and are discussed as follows:

4.1 Potential and Bond Lengths

The potential energy $(V(r))$, bond length (r_1), and (r_2) were plotted on the same graph. The graphs align with theoretical predictions. The figure illustrates the

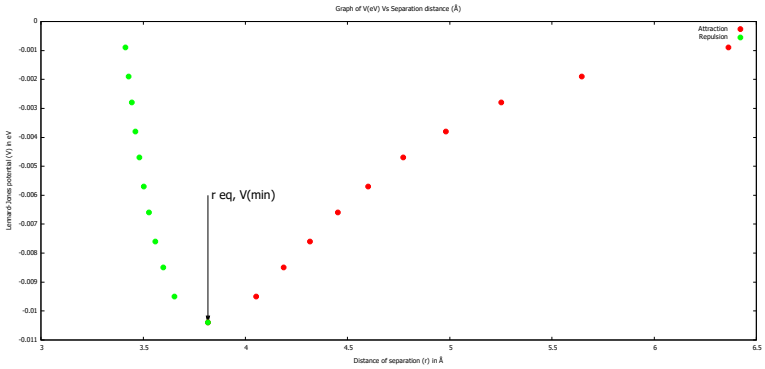

Figure 1. Potential Energy vs. Bond Length

Lennard-Jones potential, where the potential energy decreases as the interatomic distance approaches the equilibrium bond length (r_{eq}), reaching a minimum at stability. For distances less than r_{eq}, repulsive forces dominate, causing a sharp increase in energy. Beyond r_{eq}, the energy gradually rises due to weaker attractive forces and asymptotes to zero at large separations.

4.2 Force and Bond Lengths

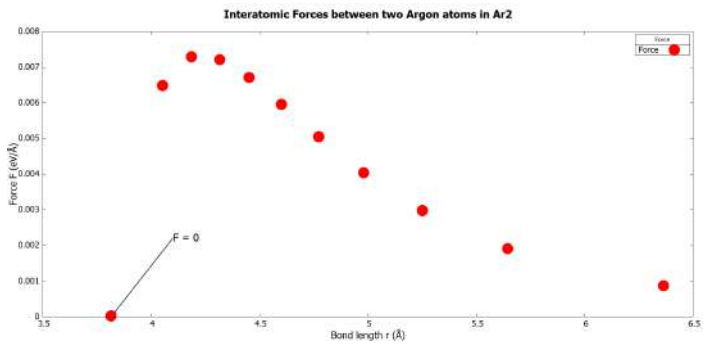

Figure 2. Force vs. Bond Length

The figure illustrates the Lennard-Jones potential, where the potential energy decreases as the interatomic distance approaches the equilibrium bond length (r_{eq}), reaching a minimum at stability. For distances less than r_{eq}, repulsive forces dominate, causing a sharp increase in energy. Beyond r_{eq}, the energy gradually rises due to weaker attractive forces and asymptotes to zero at large separations.

4.3 Dispersion Curve

The relationship between frequency (f) and energy was obtained, ensuring that $f \leq 26\,\text{cm}^{-1}$.

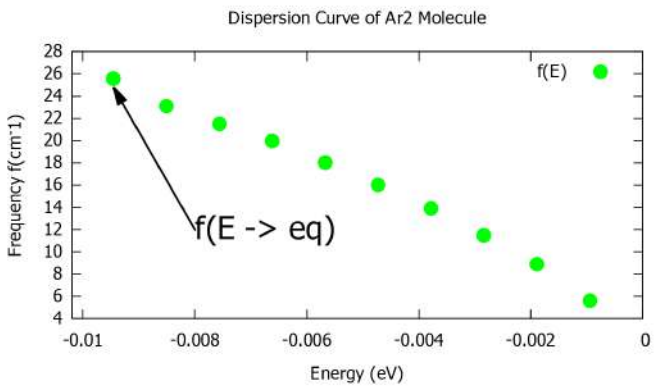

Figure 3. Dispersion Curve

5 Conclusion

This study successfully analyzed the interatomic interactions between two argon atoms using the Lennard-Jones potential through both analytical derivations and molecular dynamics simulations with LAMMPS. The findings demonstrate that the equilibrium bond length and dissociation energy are key factors in understanding atomic-scale interactions. The observed inverse relationship between energy levels and oscillation frequency underscores the harmonic nature of these interactions. These results provide valuable insights into the behavior of noble gases with potential applications in molecular dynamics, material science, and chemical engineering. Further research could extend this approach to more complex systems or explore the impact of external conditions on atomic interactions.

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APPENDIX

Distance (Å)	Force (eV/Å)
3.816717	-1.8378852077652698e-09
4.186803	-0.00729052284522499
4.362929	-0.00724107270711657
4.45167	-0.006710155776218651
4.63076	-0.00595839253187853
4.77259	-0.005004983078215417
5.02113	-0.00298292820507824
5.62515	-0.00190902580587857
6.363517	-0.000570223520453236

Table 1. Distance and Force data.

Time Step	Time (s)	F (Hz)	wave no.(cm^{-1})
2722	1.26e-12	7.39021e+11	24.4804705
2603	1.32e-12	7.32671e+11	24.3914795
2688	1.43e-12	6.69265e+11	22.3550856
2705	1.45e-12	6.5986e+11	22.0499705
2715	1.58e-12	6.10859e+11	19.9620102
1433	1.8e-12	5.6498e+11	17.9304925
1604	2.04e-12	5.1246e+11	16.0600926
8802	2.4e-12	4.58665e+11	14.3068274
7716	2.68e-12	3.995e+11	11.8950201
11382	9.06e-12	1.69485e+11	5.65675907

Table 2. Time, Frequency, and Wavenumber data.

i	E_i (eV)	a	b	c	δ	x_1	x_2	r_1'	r_2'
1	-0.00945	0.0416	-0.0416	0.00945	0.00016	0.651	0.349	3.587	4.052
2	-0.00851	0.0416	-0.0416	0.00851	0.00014	0.713	0.287	3.509	4.187
3	-0.00756	0.0416	-0.0416	0.00756	0.00012	0.761	0.239	3.558	4.316
4	-0.00662	0.0416	-0.0416	0.00662	0.00009	0.803	0.197	3.479	4.452
5	-0.00568	0.0416	-0.0416	0.00568	0.00007	0.838	0.162	3.536	4.596
6	-0.00473	0.0416	-0.0416	0.00473	0.00006	0.870	0.130	3.462	4.741
7	-0.00378	0.0416	-0.0416	0.00378	0.00003	0.899	0.101	3.413	4.881
8	-0.00284	0.0416	-0.0416	0.00284	0.00002	0.926	0.074	3.412	5.099
9	-0.00189	0.0416	-0.0416	0.00189	0.00001	0.952	0.048	3.428	5.645
10	-0.00095	0.0416	-0.0416	0.00095	0.00157	0.977	0.023	3.413	6.363

Table 3. Main table data with rearranged columns.

6 Gnuplot Script for Lennard-Jones Potential Plot

The following Gnuplot script plots the Lennard-Jones potential against the separation distance:

Listing 1. Gnuplot Script for Lennard-Jones Potential

```
# Set the labels for the axes
set xlabel 'Distance of separation (r) in  '
set ylabel 'Lennard-Jones potential (V) in eV'

# Set the title of the plot
set title 'Graph of V(eV) Vs Separation distance (  )'

# Enable grid for better readability

# Enable key (legend) at the top right corner
set key right top

# Style settings for thick filled dots
set style line 1 lc rgb "red" pt 7 ps 2  # Thick red filled dots
set style line 2 lc rgb "green" pt 7 ps 2  # Thick green filled dots
set arrow from 3.816,-0.006 to 3.816,-0.0104 head filled lw 2
set label " r eq, V(min)" at 3.816,-0.006 font "Helvetica,18"

# Plot the data: filled points only
plot 'potentialbondlength.txt' using 2:1 with points ls 1 title 'Attraction', \
     'potentialbondlength.txt' using 3:1 with points ls 2 title 'Repulsion'
```

Gnuplot Script for Force Plot

The following Gnuplot script generates a plot of the interatomic forces between two Argon atoms based on data from 'ForceDistance.txt':

Listing 2. Gnuplot Script for Interatomic Forces

```
# Set graph title and axis labels with increased font sizes
set title "Interatomic Forces between two Argon atoms in Ar2" font "Helvetica-Bold,18
set xlabel "Bond length r (  )" font "Helvetica,16"
set ylabel "Force F (eV/  )" font "Helvetica,16"

# Increase font size for tick labels
```

```
set xtics font "Helvetica,14"
set ytics font "Helvetica,14"
set yrange [0:0.008]

# Set the key with red dots, inside a box, and customized font
set key inside top right font "Helvetica,14" box opaque
set key title "Force" # Display only "Force" in the legend

# Set the style for the plot with larger point size
set style line 1 linecolor rgb 'blue' pointtype 7 pointsize 5 # Blue dots for increasi
set style line 2 linecolor rgb 'red' pointtype 7 pointsize 5
# Red dots for decreasing points with larger size
set arrow from 4.1,0.0022 to 3.816,0 head filled lw 2
set label "F = 0" at 4.1,0.0022 font "Helvetica,18"

# Plot the data and add a proper key for red dots ("F(r)")
plot 'ForceDistance.txt' using 1:2 with points linestyle 1 title "Force"
```

Gnuplot Script for Dispersion Curve of Ar_2

Below is the Gnuplot script to generate the dispersion curve of the Ar_2 molecule:

Listing 3. Gnuplot Script for Dispersion Curve

```
set title "Dispersion Curve of Ar2 Molecule"
set xlabel "Energy (eV)"
set ylabel "Frequency f(cm^-1)"
set xrange [-0.01:0]
set yrange [4:28]
set ytics 4,2,28
set offsets graph 0.02,0,0,0
set key inside top right
set style line 1 lc rgb 'green' pt 7 ps 2 lw 2 # Green thick dots
set arrow from -0.008,12 to -0.009455,24.8 head filled lw 2
set label "f(E -> eq)" at -0.008,12 font "Helvetica,18"
plot 'frequency vs energy.txt' using 2:1 title "f(E)" with points linestyle 1
```

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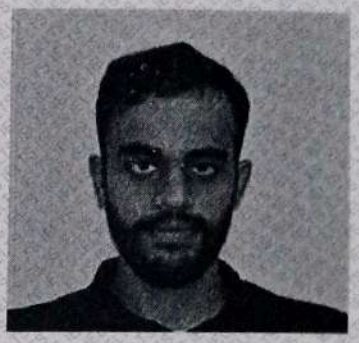

Family Name	REHMAN
First Name	SAAD UR
Candidate ID	CS5428142

Date of Birth	05/03/2001	Sex (M/F)	M	Scheme Code	Private Candidate

Country or Region of Origin	
Country of Nationality	PAKISTAN
First Language	URDU

Test Results

Listening	Reading	Writing	Speaking	Overall Band Score	CEFR Level
7.5	7.0	6.5	6.5	7.0	C1

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